

Fluid Imbibition as a Factor in Estrogen-Induced Increase of Prostatic Weight in Castrated Rats

J.J. Corrales, N. Kadohama, L.S. Chai, P.A. Hoisaeter, M.T. Hampton, G.P. Murphy, and A.A. Sandberg

Roswell Park Memorial Institute, Buffalo

The mechanism by which estrogens inhibit castration atrophy has been investigated morphologically and biochemically utilizing ventral prostate from Copenhagen rats. The suppression of weight loss and gross edematous appearance of the prostate associated with the *in vivo* effect of 17 β -estradiol (E₂) could not be accounted for by DNA and protein synthesis. Increase in the fluid content in the tissues was confirmed by demonstration of significant increase in the ratio of wet/dry tissue weights. Light microscopy demonstrated that the main effects were on the stroma, characterized by large interglandular areas almost totally devoid of collagen resulting in an edematous appearance. Electron microscope studies showed an abundance of fluid localized adjacent to the capillary endothelium and some red blood cells indicating disturbances in capillary permeability. The combination of a prolactin secretion inhibiting agent with E₂ partially prevented the fluid accumulation and weight increase produced by E₂ alone, indicating an involvement of prolactin in the estrogen effect. Differences in blood prolactin concentration between strains of rats may influence the sensitivity of the prostate to estrogens.

Key words: prostate, estrogens, prolactin, organ culture, morphology

INTRODUCTION

Even though the effects of estrogens on the prostate of castrated rats have been recognized for decades [1], the hormonal mechanism of action, as well as the morphological aspects of these effects, are far from clear.

The results describing the morphological aspects of the estrogen effect are conflicting. Some authors have found the main effect at the fibromuscular level [2, 3], whereas others indicate an exclusive action on the epithelium and its secretion [4, 5]. In this respect, we previously observed that the prostatic tissue of the Copenhagen rats appeared edematous after E₂ treatment, possibly indicating changes in the relative distribution of fluids and electrolytes within the prostate and/or changes in the secretory activity [6]. Therefore, we have studied in our model the estrogen effects on the histological and ultrastructural appearance of prostatic tissue as well as on some biochemical parameters. Also reported herein are the histological effects on the prostate of elymoclavine, an inhibitor of prolactin secretion [7, 8], when administered alone or in combination with estrogens. Information in the literature on this point is, at best, scarce.

Address reprint requests to Dr. Avery A. Sandberg, Roswell Park Memorial Institute, 666 Elm Street, Buffalo, NY 14263.

The ability of estrogens to inhibit castration atrophy has mainly been explained as a direct effect of the estrogens on the prostate [9]. These results are in line with those of our previously published report showing a direct effect of E_2 on prostatic tissue in organ culture [6]. However, this observation does not exclude the possibility that estrogens may also influence prostatic tissue indirectly. Of particular interest in this respect is prolactin, inasmuch as prolactin receptors have been found in the prostate [10, 11], and this hormone influences the androgen uptake and metabolism in the prostate [12, 13].

MATERIALS AND METHODS

Experimental Animals and Treatment Schedules

In vivo experiments. Copenhagen rats (four months old) were supplied by ARS Sprague-Dawley, Gibco Animal Resources Laboratories (Madison, Wis) and Dr. N.H. Altman of Papanicolaou Cancer Research Institute (Miami, Fla). Wistar-Furth rats (six months old) were purchased from Microbiological Associates, Inc (Bethesda, Md). The rats were fed a standard diet (Teklad Pellets, Teklad Laboratory Animal Diets, Windfield, Iowa), given water ad libitum and maintained under standard conditions. All the animals employed for in vivo experiments were castrated via the scrotal route under ether anesthesia. Twenty-four hours after orchiectomy treatment was carried out as follows (Tables I and II):

Group A: Nine Copenhagen rats treated with peanut oil.

Group B: Nine Copenhagen rats treated with E_2 in peanut oil.

Group C: Five Wistar-Furth rats treated with peanut oil.

Group D: Five Wistar-Furth rats treated with E_2 in peanut oil.

Group E: Three Copenhagen rats treated with peanut oil and 10% ethanol in saline.

Group F: Eight Copenhagen rats treated with E_2 in peanut oil and 10% ethanol in saline.

Group G: Eight Copenhagen rats treated with E_2 in peanut oil and elymoclavine in 10% ethanol in saline.

Group H: Three Copenhagen rats treated with elymoclavine in 10% ethanol in saline.

Treatment was continued for seven days by daily injections. The animals were given 50 μ g of E_2 subcutaneously in 0.1 ml of peanut oil with or without 100 μ g of elymoclavine intraperitoneally in 0.5 ml of 10% ethanol in saline. The control rats received only 0.1 ml of peanut oil with or without 0.5 ml of 10% ethanol in saline. In the rats receiving elymoclavine or its vehicle, the last injection was performed one hour before sacrifice. The remaining rats were sacrificed by cervical dislocation under ether anesthesia. In the remaining groups, after light anesthesia, 6 ml of blood was obtained from the heart for prolactin determination. The ventral and dorsolateral prostates were dissected out and weighed.

For the experiment to determine wet weight/dry weight ratios (Table III), groups A, B, C, and D were repeated, except that five rats were used for each group. Ventral and dorsolateral lobes were weighed immediately after dissection. Dry weights were determined by placing the tissues in an oven maintained at 105°C. Weights were determined every half hour for two hours. Constant dry weight was usually attained after the first half hour.

For serum prolactin measurements in different strains of rats, six Wistar-Furth and five Copenhagen rats (all four months old) were castrated and 24 hours later started on daily injections of 0.1 ml of peanut oil subcutaneously (SC) and 0.5 ml of 10% ethanol in saline intraperitoneally (IP). This treatment was carried out for seven days. The last injection of 10% ethanol was one hour before obtaining blood via cardiac puncture for prolactin measurements under the same conditions as before.

In vitro experiments. For the in vitro experiments, six Copenhagen rats (four months old) and ten Wistar-Furth rats (six months old) were castrated 24 hours before they were sacrificed. The ventral and dorsolateral prostates were removed and organ culture was performed as described previously [14]. Explants prepared from the ventral and dorsolateral lobes were maintained in culture for seven days in medium-199 containing varying concentrations of E_2 (0 to 0.1 μM). The medium was changed every two days. For each sample, weights were determined on pooled explants (5–7 pieces, each approximately 2 mm³). For weight evaluations, the combined explants were first carefully absorbed of excess medium with filter paper. Pieces of tissue were transferred to Bouin's solution for histology.

In some experiments, explants from ventral prostates of noncastrated Copenhagen rats were cultured for 0, 2, 4, and 6 days in medium-199 containing E_2 (0 to 0.1 μM) and 0.1 μM of ³H-thymidine, as detailed in "DNA Synthesis in Culture."

Combined in vivo-in vitro experiments. Three Copenhagen rats (four months old) were castrated and, after 24 hours, treated for seven days with daily injections (subcutaneous) of E_2 (50 μg) in 0.1 ml of peanut oil. For in vitro experiments, the prostates of two of the rats were cultured with ³H-thymidine and ¹⁴C-amino acids (Table VI). The remaining rat of this group received 0.8 ml IP of ¹⁴C-amino acids (7 μCi /100 g body weight) one hour before sacrifice (in vivo experiments, Table VI).

As a control, four rats were castrated and received daily injections of peanut oil. In vivo and in vitro experiments were performed in parallel with the treated group described above and also on a third intact and nontreated group of animals (two rats).

The prostatic tissues were analyzed for DNA and protein content and ¹⁴C-amino acid uptake (in vivo). In vitro incorporation of ³H-thymidine and ¹⁴C-amino acids into DNA and protein, respectively, was used to determine the capacity of the tissues to synthesize these macromolecules. Incubation for one hour in Krebs-Ringer phosphate buffer, pH 7.0, containing ³H-thymidine and ¹⁴C-amino acids at 37°C in a shaking water bath was performed (as indicated in Table VI) and the radioactivity incorporated measured.

DNA Synthesis in Culture

For the determination of ³H-thymidine incorporation as a measure of DNA synthesis in culture, explants were transferred from nonradioactive media to those containing ³H-thymidine at a final concentration of 0.2 $\mu Ci/ml$. The explants were exposed to radioactive thymidine for five hours, after which they were pooled, weighed, homogenized, and the proteins precipitated with cold 10% TCA, followed by two washes with 5% TCA. DNA was determined as described below.

Biochemical Determinations

DNA measurements were made of the supernatants of 0.3 M perchloric acid hydrolyzates according to Burton's method [15]. Protein measurements were done using the Lowry method [16]. Serum samples were assayed for prolactin using the NIAMD kit, and results expressed as nanograms of NIAMD-prolactin-PR-1/ml of serum.

Histology

For light microscopic studies the prostatic specimens were fixed in Bouin's solution, embedded in paraffin, and stained with hematoxylin-eosin or Masson's trichrome.

Electron Microscopy

For fine structure studies, tissues from groups A and B were dissected into small pieces and immediately fixed with 3% glutaraldehyde in Millonig phosphate buffer [17] at pH 7.3 for 30 minutes. The tissues were washed once for 30 minutes with the same buffer and fixed with 1% osmium tetroxide for one hour. The specimens were subsequently dehydrated in graded ethyl alcohol and propylene oxide, left in epon overnight and embedded. Sections were cut with an LKB ultra microtome and stained with uranyl acetate and lead citrate [18]. Photographs were taken with a JEM-7 electron microscope at an accelerating voltage of 80 kV.

Chemicals

Elymoclavine (8,9-didehydro-6-methylergoline-8-methanol) was kindly supplied by Dr. D.C. Kornfeld of Lilly Research Laboratories (Indianapolis, Ind). E_2 was purchased from Steraloids, Inc (Wilton, NH). 3H -[methyl,1' $^2'$]-thymidine (S.A. 95.4 Ci/mmol) and ^{14}C -L-amino acid mixture were purchased from the New England Nuclear Corp (Boston, Mass). Medium-199 was obtained from Grand Island Biologicals (Grand Island, NY).

RESULTS

Ventral and Dorsolateral Prostate Weights in the In Vivo Experiments

In Table I are presented the E_2 effects on body and prostatic weights. There were no differences in the body weights of the Copenhagen or Wistar-Furth rats treated with peanut oil versus those treated with E_2 . In the Copenhagen rats, the mean ventral prostate weight per 100 g of body weight was significantly higher in the E_2 -treated than in the peanut oil-treated groups ($P < 0.0001$). Also, a smaller but statistically significant difference was observed in the dorsolateral prostates of these rats ($P < 0.05$). In the Wistar-Furth rats, no significant differences in the mean ventral and dorsolateral prostate weights per 100 g of body weight were observed between the peanut oil-treated and the E_2 -treated animals.

The results of elymoclavine treatment on the E_2 effects on ventral and dorsolateral prostate weights are shown in Table II. Once more, the corrected weight of the ventral prostate was higher in the rats treated with E_2 than in those treated

TABLE I. Mean Body Weights and Mean Weights of the Ventral (VP) and Dorsolateral (DLP) Prostates per 100 Grams of Body Weight in Wistar-Furth and Copenhagen Rats Treated for Seven Days With Peanut Oil or Estradiol in Peanut Oil

Rat group	No. of rats	Body wt. (g) at start of treatment	VP (mg)	DLP (mg)	VP wt. (mg) per 100 g of body wt.	DLP wt. (mg) per 100 g of body wt.
Copenhagen						
Group A (peanut oil)	9	262 ± 26	52 ± 13	81 ± 41	19.9 ± 4.6	35.3 ± 11.4
Group B (estradiol)	9	261 ± 18	111 ± 25	119 ± 30	42.2 ± 9.4 ^a	45.0 ± 8.4 ^b
Wistar-Furth						
Group C (peanut oil)	5	335 ± 12	93 ± 16	132 ± 48	27.7 ± 3.9	39.2 ± 13.4
Group D (estradiol)	5	338 ± 32	103 ± 18	132 ± 23	30.6 ± 5.0	38.9 ± 5.8

^a $P < 0.0001$.

^b $P < 0.05$

TABLE II. Effect of Prolactin Secretion Inhibition on the Response of the Ventral (VP) and Dorsolateral (DLP) Prostate Weights to Estrogens

Rat group and treatment	No. of rats	Body wt. (g) at start of treatment	VP (mg)	DLP (mg)	VP wt. (mg) per 100 g body wt.	DLP wt. (mg) per 100 g body wt.	Prolactin (ng/ml)
Group E (Peanut oil + 10% ethanol)	3	274 ± 38	87 ± 21	141 ± 30	29 ± 3.5	51 ± 4	173 ± 99
Group F (Estradiol)	8	236 ± 24	112 ± 41	126 ± 44	47 ± 14	53 ± 14	786 ± 119
Group G (Estradiol + elymoclavine)	8	229 ± 26	94 ± 21	118 ± 35	41 ± 9	51 ± 13	312 ± 130
Group H (Elymoclavine)	3	273 ± 16	75 ± 11	126 ± 21	28 ± 4	46 ± 5	9 ± 5.3

with ethanol-peanut oil only. The combination of E₂ *plus* elymoclavine produced a lower weight in the ventral prostate than E₂ alone. Treatment with elymoclavine alone had no effect on the ventral prostate, as shown by the similarity of the corrected weights when compared with the control animals (Group E). No effect on the dorsolateral prostate could be observed when elymoclavine alone or combined with E₂ was administered.

The prolactin values after treatment with E₂ and/or elymoclavine are also shown in Table II. The maximum values were observed with the E₂ alone. The elymoclavine when combined with E₂ partially suppressed the increase in weight produced by the E₂, though the values were still higher than in the control group. Elymoclavine given alone led to the lowest values. There was a high degree of correlation between prostatic weights (uncorrected) and serum concentrations of prolactin ($r=0.987$, $P < 0.001$) among the four groups. The correlation coefficient was 0.923, $P < 0.01$ after normalization of prostatic weights.

In light of the unusually high control levels of prolactin (Table II), it was necessary to perform measurements under the same conditions in a second strain

TABLE III. Effect of Estradiol on the Wet and Dry Weights of Prostates of Copenhagen and Wistar-Furth Rats*

Rat group	Peanut oil	Estradiol	% Increase
Copenhagen			
VP Wet wt.	38.2 ± 3.0 ^a	61.4 ± 2.5 ($P < 0.005$)	60
Dry wt.	16.0 ± 0.6	20.1 ± 1.0	25
Ratio wet wt./dry wt.	2.38	3.05	
DLP Wet wt.	41.9 ± 0.3	55.1 ± 2.6 ($P < 0.005$)	31
Dry wt.	16.4 ± 0.3	20.3 ± 0.9	25
Ratio wet wt./dry wt.	2.55	2.80	
Wistar-Furth			
VP Wet wt.	33.2 ± 1.8	37.4 ± 2.3 (n.s.)	12
Dry wt.	11.7 ± 0.5	12.9 ± 0.6	10
Ratio wet wt./dry wt.	2.83	2.90	
DLP Wet wt.	44.5 ± 2.9	56.0 ± 1.5 ($P < 0.01$)	25
Dry wt.	14.0 ± 0.9	16.6 ± 0.4	18
Ratio wet wt./dry wt.	3.17	3.37	

*Animal groups as well as the treatments were as in the experiment presented in Table I.

^aProstate weights as expressed in mg/100 g body weight. All values are means of five rats ± standard errors.

of rats. Serum prolactin concentration in the Wistar-Furth rats was 20.2 ± 6.3 ng/ml (mean ± SD). In the Copenhagen strain, the value was 75.1 ± 8.8 ng/ml. There was a clear statistical difference between the serum prolactin levels in the two strains ($P < 0.0005$).

To confirm the presence of fluid as a major contributing factor in the weight differential and swollen appearance of the ventral prostates of E₂-treated Copenhagen rats, determination of ratios of wet/dry tissue weight was undertaken. As presented in Table III, a 60% increase in wet weight of ventral prostate of Copenhagen rats after E₂ administration was accompanied by 25% gain in dry weight. On the other hand, despite a 25% increase in dry weight of the dorsolateral lobe, E₂ produced only 31% increase in wet weight. Prostates of Wistar-Furth rats showed relatively small response to E₂ in terms of both wet and dry weights. Thus, the largest difference in the wet/dry weight ratio between E₂-treated and peanut oil-treated control animals was clearly in the ventral prostates of Copenhagen rats, ie, 3.05 ± 0.09 SEM ($n = 5$) vs 2.38 ± 0.10 SEM ($n = 5$), $P < 0.005$.

Ventral and Dorsolateral Prostate Weights in Organ Culture Experiments

The effects of several concentrations of E₂ on explants derived from ventral and dorsolateral prostates are shown in Table IV. Explants maintained without estrogenic steroids in the culture medium for the seven-day period were considered to be the controls.

In the Copenhagen rats, the weights of the ventral prostate explants were always higher in those cultured in E₂-supplemented media than those maintained in hormone-free media. Suppressions in loss of weight were related to E₂ concentration. Thus, the weight differential was greatest with $0.1 \mu\text{M}$ E₂ ($P < 0.01$), followed by $0.01 \mu\text{M}$ ($P < 0.02$) and least with $0.001 \mu\text{M}$ (difference not significant).

TABLE IV. Effect of Estradiol on the Weight (Mean $\pm$ SD) of Ventral (VP) and Dorsolateral (DLP) Prostate Explants* From Copenhagen and Wistar-Furth Rats

Rat group	Estradiol dosage (μ M)	Mean weight (mg/explant) ^a	
		VP	DLP
Copenhagen			
Noncultured prostates	—	2.94 $\pm$ 0.15(3) ^b	2.94 $\pm$ 0.12(3)
Cultured prostates	—	1.39 $\pm$ 0.20(4)	2.04 $\pm$ 0.16(2)
	0.001	1.61 $\pm$ 0.27(3)n.s.	1.80 $\pm$ 0.30(2)
	0.01	1.73 $\pm$ 0.07(3)p < 0.02	1.72 $\pm$ 0.10(2)
	0.1	1.85 $\pm$ 0.10(3)p < 0.01	1.92 $\pm$ 0.28(2)
Wistar-Furth			
Noncultured prostates	—	3.46 $\pm$ 0.16(4)	2.78 $\pm$ 0.40(4)
Cultured prostates	—	1.67 $\pm$ 0.23(4)n.s.	2.03 $\pm$ 0.40(3)
	0.001	1.66 $\pm$ 0.12(4)n.s.	2.21 $\pm$ 0.14(3)
	0.01	1.89 $\pm$ 0.11(4)n.s.	1.90 $\pm$ 0.25(3)
	0.1	1.83 $\pm$ 0.27(4)n.s.	1.99 $\pm$ 0.18(3)

*Cultures were continued for seven days.

^aFive to seven explants per sample.

^bNumber of determinations shown in parenthesis.

In the statistics analysis, comparisons were made between the cultured prostates in media supplemented with E₂ and E₂-free medium.

No response in weight of the dorsolateral prostate explants could be observed; in fact, a slight decrease in weight was seen.

E₂ effects on the ventral prostates of Wistar-Furth rats were smaller than those in the Copenhagen rats. No dose-response relationship was shown. In the dorsolateral prostates, only the lowest concentration of E₂ (0.001 μ M) produced a partial inhibition of the castration effect of the culture explants without hormones.

Sequential Effects of Estradiol on Ventral Prostate Weights and DNA Synthesis in Organ Culture

In Table V are shown the effects of 0.01 and 0.1 μ M of E₂ on the weight of explants derived from the ventral prostates of Copenhagen rats. The main loss of weight was always observed in the first two days of the culture period. Subsequently, the losses were smaller, especially when the medium was supplemented with higher concentration of E₂ (0.1 μ M). Assigning 100% to the average weight of the three samples at day 0 (4.87 mg/explant) and subtracting from this figure the weights recorded on subsequent days, the following weight losses were recorded: 41%, 51%, and 57% at days 2, 4, and 6, respectively, in the control medium; corresponding figures in 0.01 μ M E₂ were 29%, 46%, and 51%; in 0.1 μ M E₂, 38%, 41%, and 42%. Thus, there was an inverse relationship between the weight loss and the E₂ concentration in the medium.

The incorporation of ³H-thymidine as a measure of DNA synthesis showed a progressive increase with time, despite clear weight losses in the explants. The control medium and 0.01 μ M E₂ were associated with similar rates of incorporation that peaked at day 4. In 0.1 μ M E₂, an early effect was suppression of thymidine uptake with respect to the control. However, an increase in the incorporation was observed by day 6, indicating stimulation of these explants into active DNA synthesis.

TABLE V. Effect of Estradiol on Weight and DNA Synthesis in Explants Derived From Ventral Prostates of Intact Copenhagen Rats

Days	Control medium		0.01 μ M estradiol		0.1 μ M estradiol	
	mg/explant ^a	DPM/30mg ^b	mg/explant	DPM/30 mg	mg/explant	DPM/30 mg
0	4.87	460	4.87	877	4.87	252
2	2.88	3,765	3.46	4,390	3.00	1,408
4	2.37	5,362	2.59	5,766	2.85	1,141
6	2.08	4,339	2.37	4,113	2.80	4,271

^aEach value represents a single determination from a total of ten explants per sample.

^bAfter four hours exposure to medium containing 0.1 μ M of ³H-thymidine.

Changes in DNA and Protein Content and Synthesis Induced by Estradiol

In vivo experiments. In the upper part of Table VI are shown the effects of E₂ and castration on DNA and protein content and synthesis in the rat ventral prostate. As expected, the ventral prostate weight from the E₂-treated animal was higher than in the peanut oil-treated control but still lower than in the intact animal. The highest amount of DNA, both in absolute and relative values, was obtained in the peanut oil-treated rats. Despite a threefold greater mass of tissue, the DNA content (per mg of tissue) in ventral prostates of E₂-treated rats was less than 20% of that of peanut oil-treated control samples, though slightly higher than intact values. Similarly, the relative protein content (expressed per mg of tissue) was the lowest in prostates of E₂-treated rats.

New protein synthesis, as judged by the ¹⁴C-amino acids uptake, was much lower in both castrated and E₂-treated rats than in intact animals. Even though the E₂-treated rat showed a lower uptake than the castrated controls when the radioactivity was related to tissue weight, the values were higher when related to mg of protein. The correct relative value was related to per mg of protein, taking into account the fact that water imbibition explains, at least partially, the differences in prostate weight between the castrated groups.

In vitro experiments. The results are presented in the lower part of Table VI. The DNA and protein values observed in the in vitro experiments followed the same trend as in the in vivo experiments.

The capacity for DNA synthesis, based on thymidine incorporation in incubated minces, showed the highest values for the intact animals. The levels observed in tissues from the E₂-treated rats (expressed per 100 μ g of DNA) were, in turn, higher than those in the castrated control group. Even though the DNA content was much lower, tissues derived from estrogenized rats obviously retained a greater potential to synthesize DNA than that of control animals, further indicating factors other than regeneration through macromolecular synthesis as primarily accountable for the estrogen effect.

Protein synthesis studies showed that the minces derived from the E₂-treated animals incorporated higher amounts of radioactivity than those derived from either the peanut oil-treated or the intact rats. As in the in vivo experiments, the lowest levels were measured in the peanut oil-treated group. Again, despite a twofold greater capacity to synthesize proteins over control values, estrogen treatment produced an increase in protein synthesis of less than 20% in vivo.

TABLE VI. Changes in DNA and Protein Content and Synthesis Induced by Castration and Estradiol Treatment

In vivo ^a Treatment	mg tissue per ventral prostate	mg prostate tissue per 100 gm body wt.	μ g DNA in tissue	μ g DNA per mg tissue	μ g DNA per mg tissue	mg protein in tissue	100 μ g protein per mg tissue	DPM's ¹⁴ C-AAAs incorp'd	DPM's per 100 mg tissue	DPM's per mg protein
E ₂ ^b	200	85	236	1.18	1.18	7.40	3.70	8,442	4,221	1,140
PO ^c	55	27	363	6.60	6.60	3.73	6.78	3,622	6,585	970
Intact	300	129	306	1.02	1.02	15.75	5.25	23,390	7,796	1,485
In vitro ^d										
Sample No.	mg tissue per incubation	μ g DNA in tissue	μ g DNA per mg tissue	DPM's ³ H-Tdr incorp'd	DPM's per 100 μ g DNA	mg protein in tissue	100 μ g protein per mg tissue	DPM's ¹⁴ C-AAAs incorp'd	DPM's per 100 mg tissue	DPM's per mg protein
1	65	88	1.35	51,939	59,021	3.40	5.23	—	—	—
2	95	211	2.22	83,818	39,724	5.67	5.96	—	—	—
3	151	162	1.07	167,831	103,599	8.47	5.61	—	—	—
4	46	79	1.71	—	—	1.91	4.15	62,476	135,817	32,710
5	75	226	3.00	—	—	5.84	7.78	93,136	124,181	15,947
6	151	140	0.92	—	—	6.84	4.53	143,700	95,165	21,008

^aIn vivo samples were whole prostates from rats injected with ¹⁴C-amino acids (7 μ Ci/100 g body weight).

^bE₂ = 50 μ g E₂/day administered to castrated rats.

^cPO = peanut oil (0.1 ml) administered to castrated rats.

^dIn vitro samples #1 and 4 were of tissues derived from E₂-treated rats. In vitro samples #2 and 5 were of tissues derived from PO-treated rats. In vitro samples #3 and 6 were of tissues derived from intact rats.

In vitro samples #1-3 were incubated with 4 μ Ci/ml of ³H-thymidine. In vitro samples #4-6 were incubated with 0.8 μ Ci/ml of ¹⁴C-amino acids. Three rats were employed for in vivo experiments and two for in vitro studies.

Light Microscopic Studies

In vivo effects of estradiol on Copenhagen rat prostate. Comparison of the ventral prostate histology of the E₂- and peanut oil-treated rats showed differences both in the stroma and glands (epithelium). In the E₂-treated rats (Fig. 1A) the stroma was looser, less cellular and more amorphous than in the latter animals (Fig. 1B); the stroma occupied large interglandular areas giving the appearance of edematous imbibition. Furthermore, tissues from the E₂-treated rats were almost totally devoid of Masson's trichrome staining (specific for collagen). In contrast, prostates from intact untreated rats (Fig. 1C) were strikingly different from those of the castrated animals with respect to the number and size of the acini. The tissue was composed predominantly of glandular acini, which were more numerous, markedly smaller in size, and separated by a thin network of stromal tissue.

In the control peanut oil-treated group (Fig. 1B) the glandular epithelium showed mainly cuboidal cells with infolding of the epithelial border. There was more optimal maintenance and more abundance of intracellular content than in the prostates of the E₂-treated animals. Tissues from E₂-treated rats showed a variable glandular pattern consisting predominantly of distended glands lined by flat epithelium and some areas of epithelial hyperplasia causing extensive infolding of the lumina.

No differences could be found in the histology of the dorsolateral prostate between the E₂- and the peanut oil-treated rats.

In vitro effects of estradiol on Copenhagen rat prostate. In contrast to the findings in the ventral prostate *in vivo*, the main effect of E₂ *in vitro* was on the epithelium (Fig. 2A). In cultures containing 0.01 μ M of E₂, the glandular lumina featured well-maintained epithelium with cylindrical cells. Many glands showed infolding into the lumina as a result of epithelial hyperplasia. In contrast, explants maintained in hormone-free media (Fig. 2B) possessed lumina that were composed predominantly of cuboidal epithelial cells. Treatment with 0.01 μ M of E₂ was more effective than higher (0.1 μ M) or lower (0.001 μ M) concentrations used in terms of maintaining epithelium histology. No difference was recognizable in the prostatic stroma between control nonhormone and E₂ supplemented medium.

In vivo and in vitro effects of estradiol on Wistar-Furth rat prostate.

In contrast to the significant differences observed in the ventral prostate of Copenhagen rats *in vivo*, E₂ treatment produced no such differences in the stroma of the ventral prostates from Wistar-Furth rats. In the *in vitro* experiments, it was difficult to recognize differences between explants with and without supplemental hormones in the Wistar-Furth strain, in contrast to the differences observed in the prostates of Copenhagen rats. *In vitro* treatment with E₂ produced no differences in the dorsolateral prostate.

In vivo effects of elymoclavine and/or estradiol on Copenhagen rat prostate. Light microscopy of prostates from the E₂-treated and control groups was described previously. When elymoclavine was combined with E₂ (Fig. 3), the effect of the E₂ on the stroma was no longer observed; neither could areas with epithelial hyperplasia be found. The stroma was clearly stained by Masson's trichrome, in contrast to the poor staining observed when employing E₂ alone. In the ventral prostate of animals treated with elymoclavine alone, a clear resem-

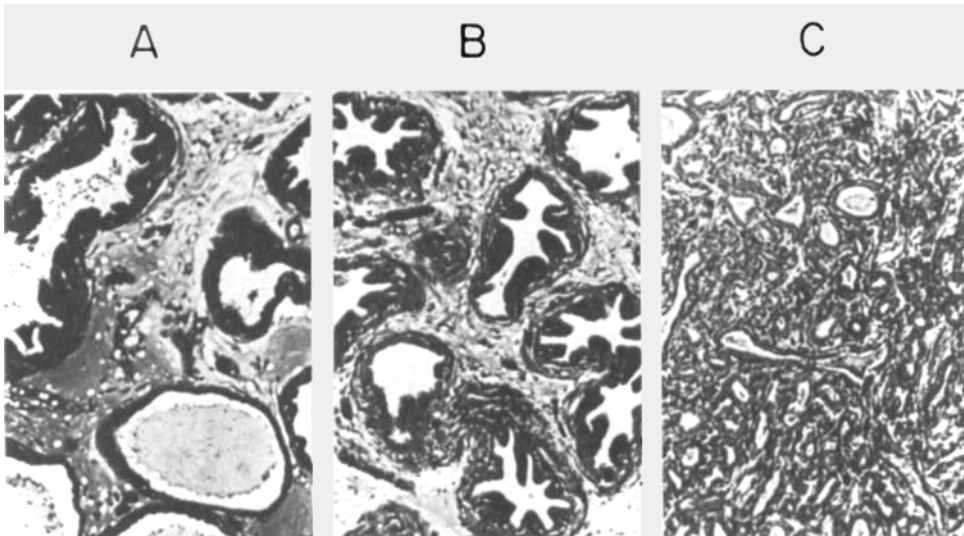


Fig. 1. Ventral prostate from Copenhagen rats (A) castrated and treated with estradiol for seven days, (B) castrated and given peanut oil for seven days, and (C) untreated. Masson's trichrome (240 $\times$).

blance to tissue from castrated controls was observed with the exception that the stroma was more abundant and more clearly stained with Masson's trichrome.

No differences could be observed in either the stroma or the epithelium of the dorsolateral prostate of control and E_2 -treated rats. However, elymoclavine alone produced marked effects in both the glandular epithelium and the fibromuscular stroma of the dorsolateral lobe. There was pronounced infolding of the glands, due to crowding of the tall columnar cells. Some areas displayed hyperplasia of the basal cells. The stroma appeared relatively dense and fibrous.

The overall picture in the dorsolateral prostate in elymoclavine plus E_2 -treated rats was intermediate between that of the E_2 - and elymoclavine-treated animals.

Electron Microscopic Studies

There were two distinct differences between the ventral prostates from peanut oil-treated (control) and E_2 -treated rats. The epithelial cells of the former were mainly of a short columnar orientation with the nuclei occupying the center of the cells (Fig. 4A). Endoplasmic reticulum was sparse and mitochondria were located mostly in the apical region. The connective tissue and muscular stroma appeared intact. In contrast, the epithelial cells from the E_2 -treated animals were squamous (Fig. 4B). Endoplasmic reticulum was also sparse, and mitochondria were scattered throughout the cytoplasm. There was an abundance of free polyribosomes in the treated cells (Fig. 4B). The Golgi apparatus in the E_2 - and the peanut oil-treated groups were laterally displaced with respect to the nucleus from their normal location in the apical region in the noncastrated animals. No differences were observed in the luminal membrane between the E_2 - and the peanut oil-treated rats or in the electron dense bodies noted in the epithelial cells of the two groups. Electron dense bodies were present in higher amounts in basal cells.

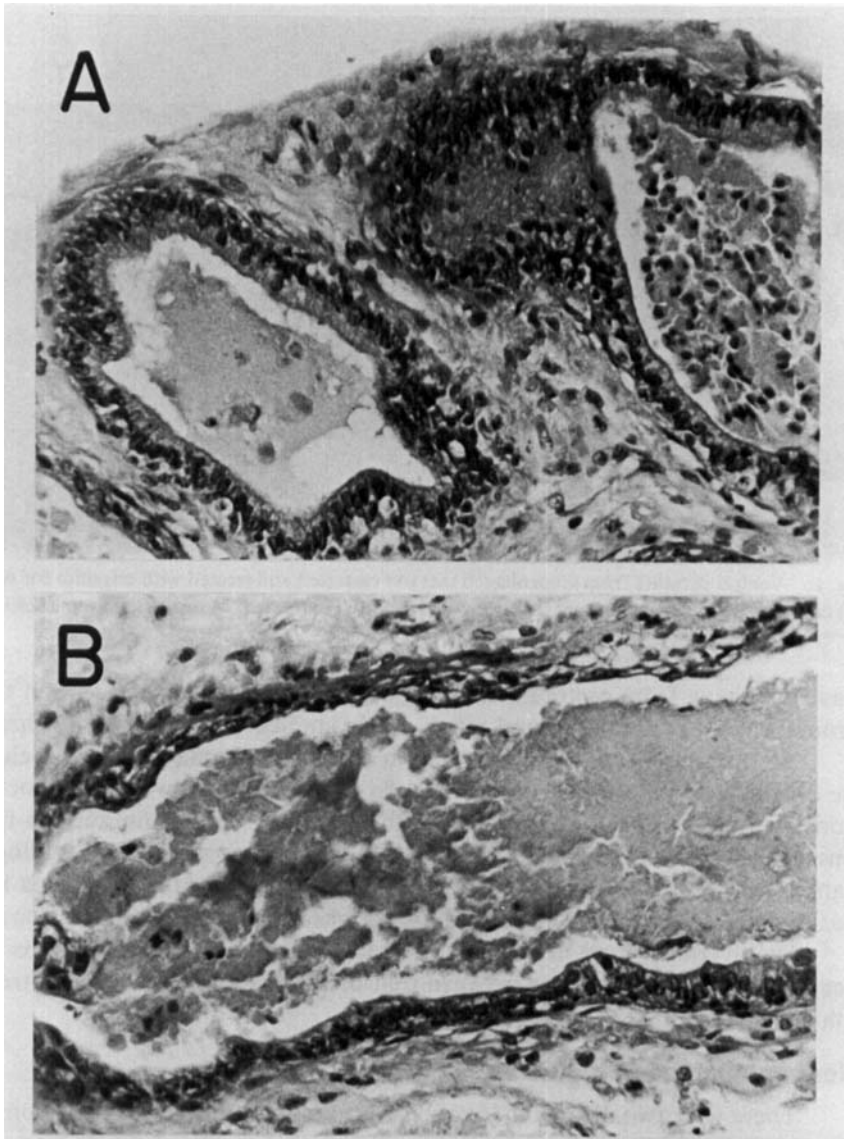


Fig. 2 Explants from ventral prostate from Copenhagen rats maintained in (A) medium containing 0.01 μ M estradiol, (B) hormone-free medium. Hematoxylin-eosine (280 $\times$).

Whereas the connective tissue and muscle cells appeared intact near the glandular epithelium (E_2 -treated rats), the fibromuscular stroma near the endothelium appeared dilated (Fig. 5A), contained irregular masses (Fig. 5B) and was characterized by complete disorientation of the collagen fibers. The processes of the fibroblasts also showed a complete structural disorientation. Even more striking was the abundance of spacing between the collagen fibers and fibroblast processes indicating an edematous condition, changes not observed in the Wistar-Furth rat prostate. Also present were white blood cells located in the edematous region but un-

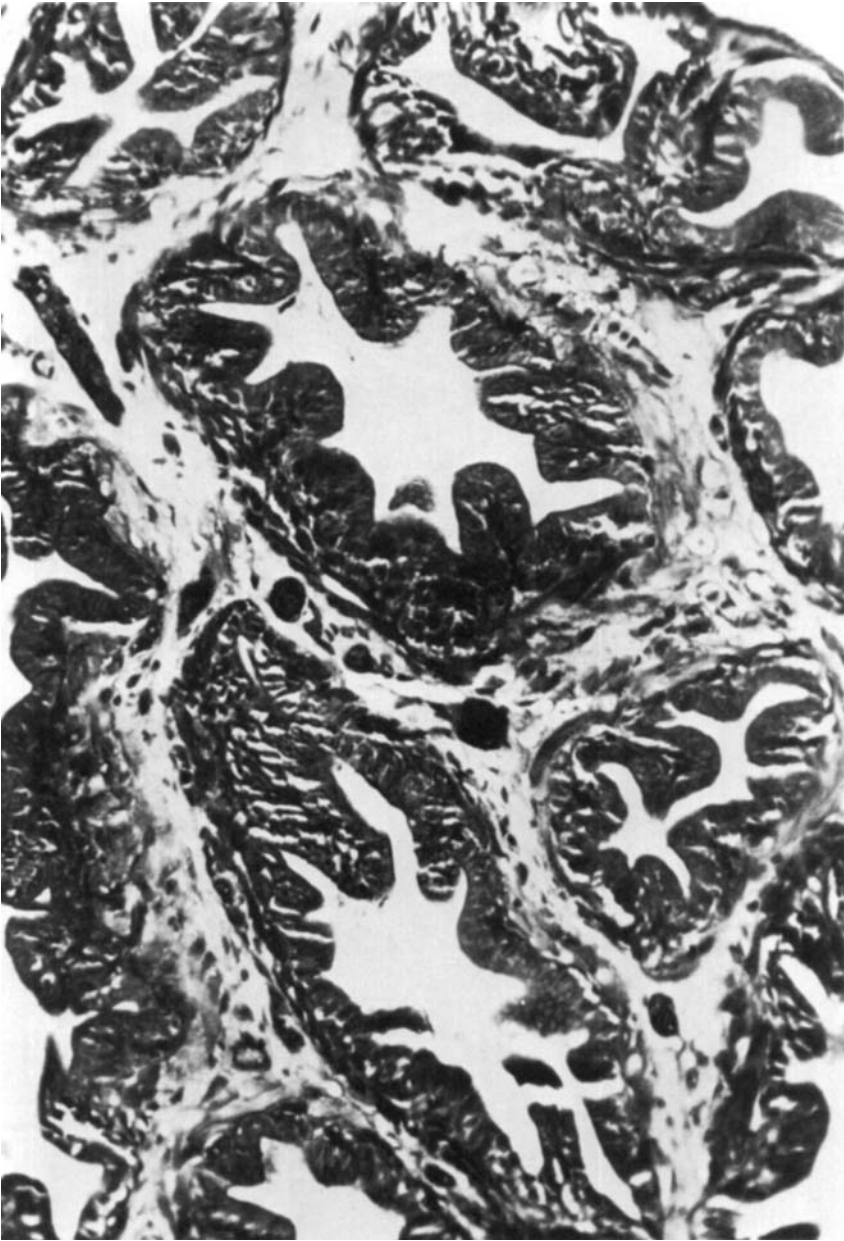


Fig. 3. Ventral prostate from rats treated with a combination of estradiol and prolactin secretion inhibiting agent, Masson's trichrome (240 $\times$).

detectable in the control tissues. The number of capillaries was greater and their size bigger in the ventral than in the dorsolateral prostate (Fig. 6).

Electron microscopy of the dorsolateral prostates showed interesting differences not readily observed with the light microscope. Whereas the epithelium in

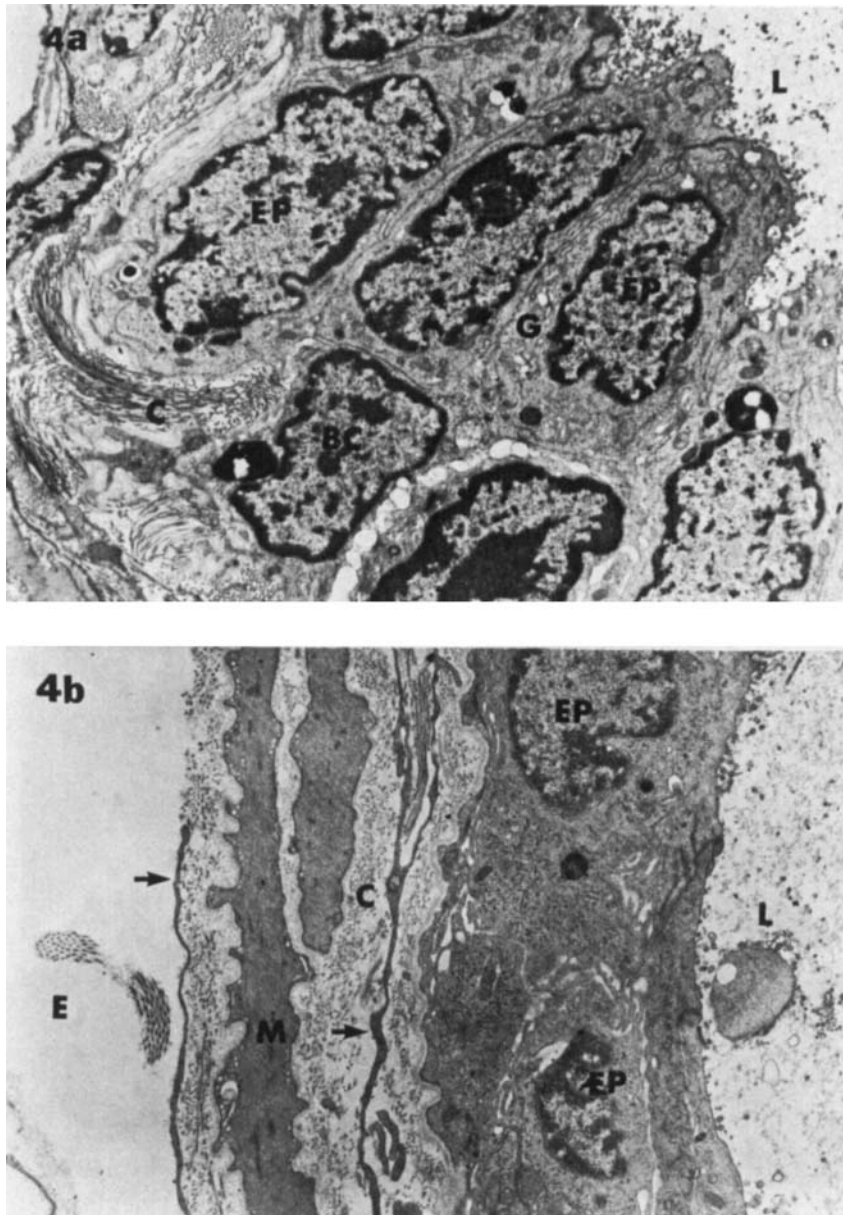


Fig. 4. Ventral prostate from castrated Copenhagen rats following (A) peanut oil injection and (B) treatment with estradiol (E_2). Changes in size and orientation of epithelial cells are presented. EP = epithelial cell; BC = basal cell; L = lumen of acinus; G = golgi apparatus; C = collagen fibers; M = muscle cells; E = edematous region; arrows = fibrocytic processes. (12,000 $\times$).

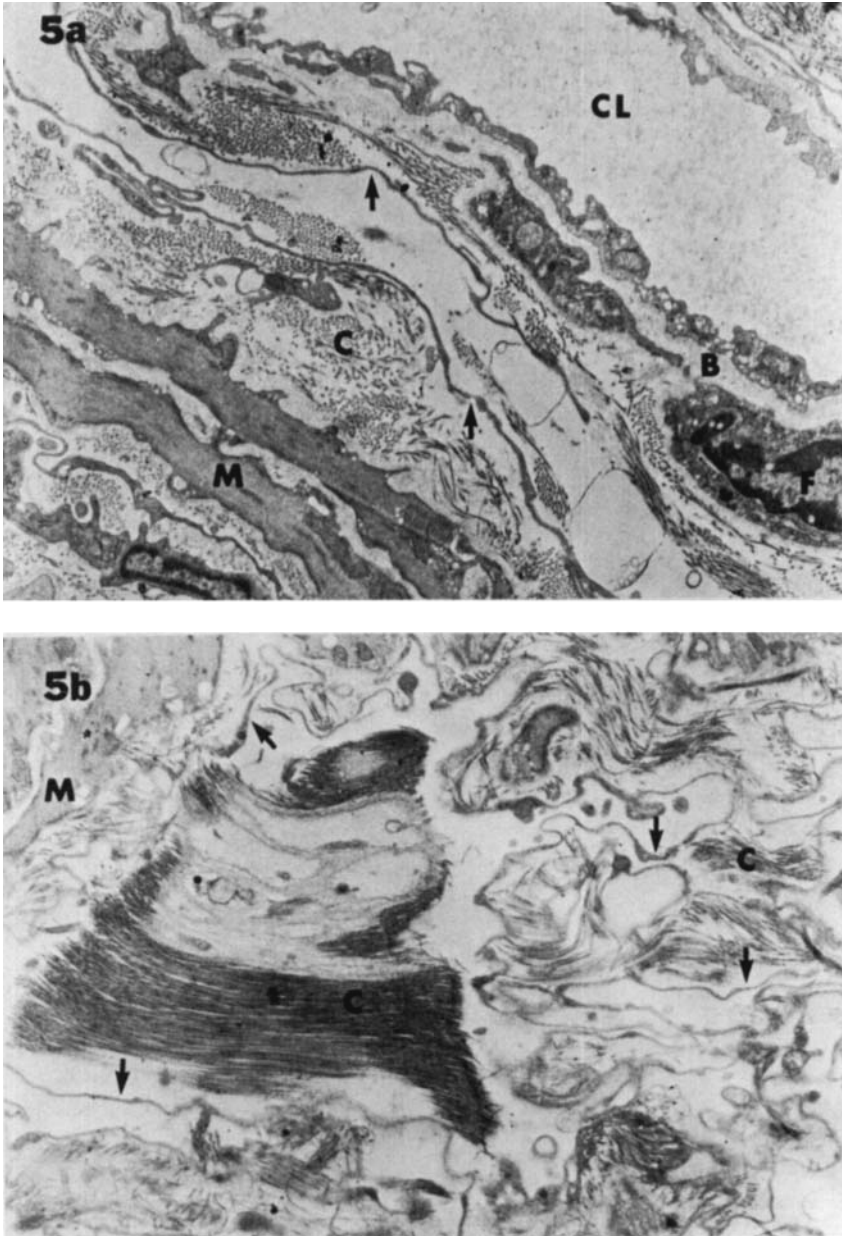


Fig. 5. Fibromuscular stroma in ventral prostate from E₂-treated rats. Edematous effect, localized near the capillaries, is shown. CL = capillary lumen; B = basement membrane; C = collagen fibers; M = muscle cell; F = fibrocytes; arrows = fibrocytic processes. Magnification: (A) 14,000 $\times$ and (B)

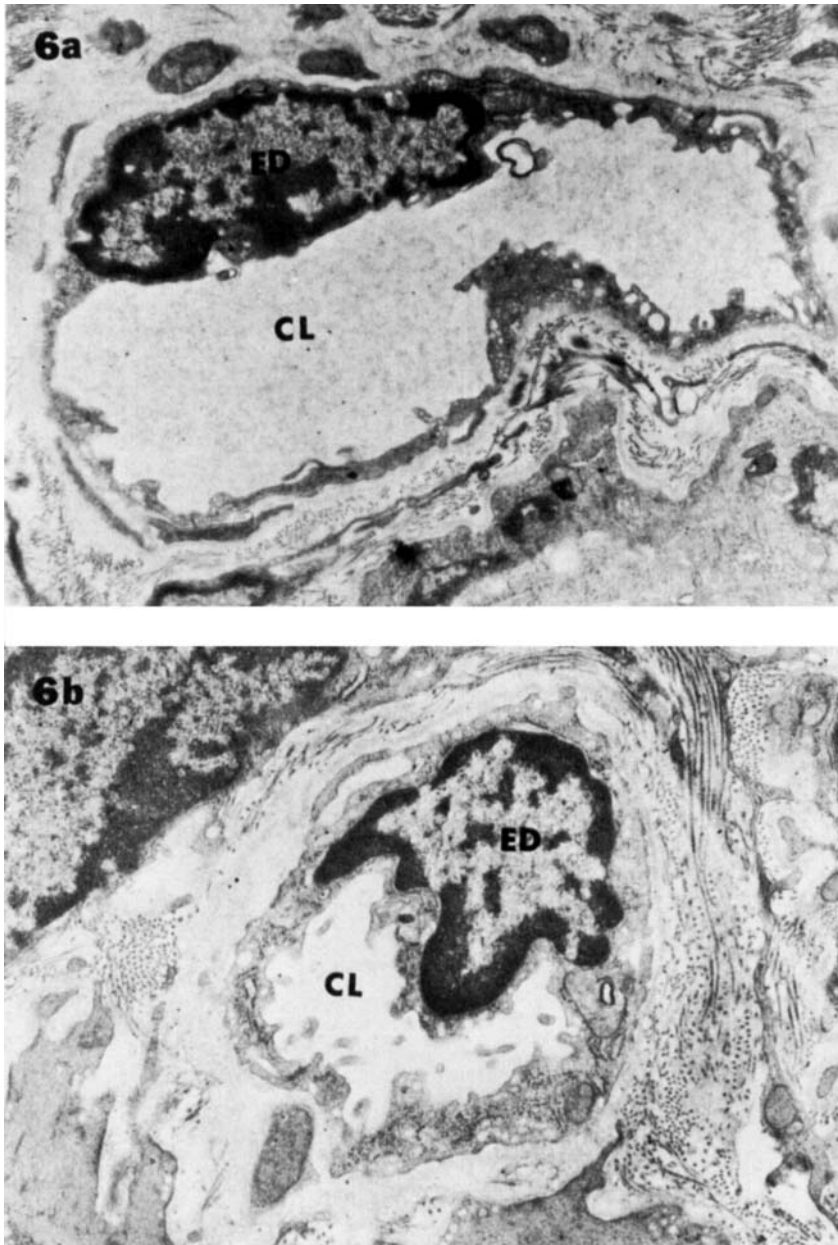


Fig. 6. Ventral (A) and dorsolateral (B) prostates from E_2 -treated rats. Obvious differences in capillary sizes are presented. Magnification: (A) 14,000 $\times$ and (B) 20,000 $\times$. ED = endothelial cell; CL = capillary lumen.

both peanut oil- and E_2 -treated groups remained cuboidal, the majority of cells of the E_2 -treated animals were extremely dilated and possessed edematous endoplasmic reticulum (Fig. 7). The connective tissue was dilated as in the ventral pro-

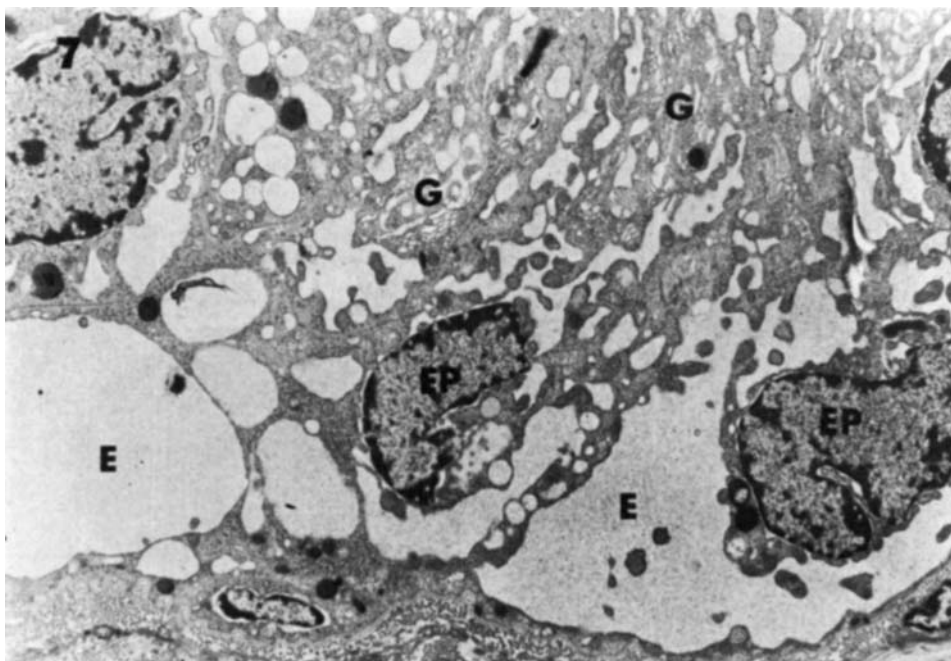


Fig. 7. Dorsolateral prostate from E_2 -treated rats. Dilatation and edematization of epithelial cells are shown. EP = epithelial cell; E = edematous region; G = golgi apparatus. (8,000 $\times$).

state (near the endothelium) and contained large masses of electron dense inclusion bodies. Occasional red blood cells were observed in the fibromuscular stroma of the E_2 -treated rats (Fig. 8) but not in that of the control group.

DISCUSSION

The morphological basis for the increase in prostatic weight induced by estrogens in castrated rats is still not unequivocally established. Recently, it has been reported that this effect involves the epithelium [4, 5]; however, other results point toward fibromuscular hypertrophy [2, 3]. Nevertheless, to the best of our knowledge, no evidence has been presented on fluid imbibition as being a causal factor.

The prominent edematous change of the ventral prostate observed macroscopically in the present study in young Copenhagen rats treated with estrogens *in vivo* suggests an increase in net content of fluid. This edematous change was associated with significant increases in prostatic weight indicating a causal relationship. The difference in prostatic weights was only partially accounted for by an increase in dry weights of the tissue, confirming that fluid imbibition is a major contributing factor. The morphological and biochemical results presented here cast some light on the origin of this fluid imbibition and increases in weight after estrogen treatment. Light and electron microscopic studies showed the fibromuscular stroma adjacent to the endothelium to be dilated, resulting in an abundance of spacing between collagen fibers and fibroblast processes. These

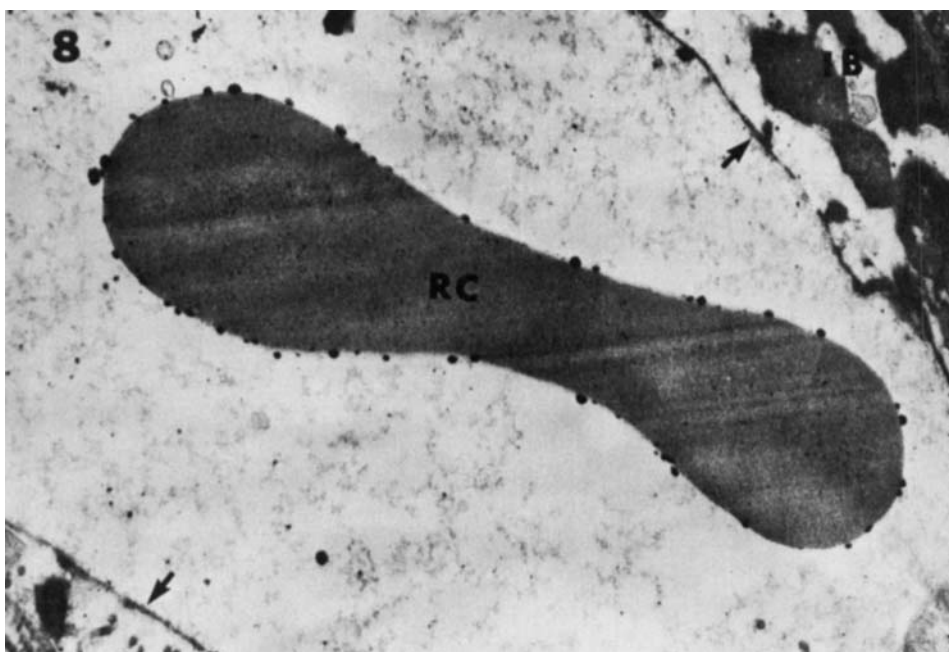


Fig. 8. Dorsolateral prostate from E₂-treated rats. Presence of red blood cells in the fibromuscular stroma is shown. RC = red blood cell; IB = inclusion bodies; arrow = fibrocytic processes. (18,000 $\times$).

changes, seen only in the ventral prostates of E₂-treated Copenhagen rats, were coincident with macroscopic edematous appearance, presence of blood elements, and localization proximal to the vascular endothelium. These observations taken together point to disturbances of the capillary permeability, which normally leads to fluid accumulation, as a possible source of edema. It is unlikely that an increase in glandular fluid accounts for the fluid accumulation, since the fluid accumulation was not intraglandular. Moreover, sites of epithelial squamous metaplasia, such as those produced by estrogens, have reduced capacity for glandular secretion [19]. Other possible factors such as the cellular elements in the stroma do not contribute to the weight increase, since the stroma in the castrated estrogen-treated animals showed less cellular content than in the rats treated only with peanut oil.

Vascular involvement in the estrogenic effects on the female sexual accessory glands is well documented [20]. Edematization mediated by an increase in the blood flow and capillary permeability constitutes an early feature in the response of these organs to estrogens [21, 22]. It has been reported that the vascular effects produced by estrogens are causally related to the ensuing hypertrophy and that the following cell division results from an increase in the availability of substrates [20]. In our model, it seems likely that the early effects of estrogens feature similar patterns as those described in the female, at least with respect to fluid extravasation.

The capillary size and number were higher in the ventral than in the dorsolateral prostates of both estrogen- and peanut oil-treated animals. This may be a

factor in the higher sensitivity to estrogens (increase in weight) of the ventral vs that of the dorsolateral prostate.

The edematization produced by E_2 was observed only *in vivo*. Its absence *in vitro* indicates that the estrogenic steroid acts in conjunction with other factors not present *in vitro*. Prolactin may constitute one of these factors, since estrogens are the most potent hormonal agents stimulating prolactin secretion in both sexes [23]. In fact, the combination of estrogen plus the antiprolactin drug, elymoclavine, suppressed the edematization effect, as well as the increase in prostatic weight to some degree, compared with the results following treatment with estrogen alone. Prolactin involvement in the estrogen effect, as in our findings, has not been reported by other studies employing antiprolactin agents [24, 25]. However, we have based our studies on a model particularly sensitive to estrogens [6]. Other workers did not employ estrogens alone [25], nor did they quantitate the prolactin blood levels after antiprolactinic agents in combination with estrogens [24, 25]. Such approaches are necessary to implicate the suppressive effects of the prolactin inhibiting secretion agents on the estrogen action. Recently, however, it was observed that increases in endogenous levels of prolactin produced by ectopic pituitary transplants led to increases in weight, both in the ventral and dorsolateral prostate in castrated and hypophysectomized rats [26].

The exact reason for the fact that edematization was only observed in the Copenhagen strain is obscure. However, this strain of rats has been found to be unique in other respects as well, eg, the spontaneous appearance of prostatic adenocarcinoma [27] and a control prolactin blood concentration in the Copenhagen castrated rat ten times higher (Table II) than that normally found in other strains [7]. Furthermore, we have observed a clearly higher prolactin blood concentration in the Copenhagen rats than in the estrogen-insensitive Wistar-Furth strain, which had normal values. It is possible, therefore, that hyperprolactinemia and/or a higher response to estrogens by the pituitary in prolactin secretion led to a higher sensitivity of the prostate in its response to estrogenic steroids. However, the isolated suppression of prolactin in the castrated rat had no influence on the prostatic weight, as indicated by the lack of differences between the control group and the antiprolactin drug group. Our results indicate that the prolactin action is additive.

Even though edematization partially explains the increase in weight induced by the estrogens in our prostatic model, a suppression in weight loss was also observed (though to a lesser degree) *in vitro* where the edematization was not seen. Furthermore, the rats treated *in vivo* with estrogens in combination with inhibitors of prolactin secretion still had higher prostatic weights than the control animals even though edematization was absent. Therefore, other factors are involved in the estrogenic effects. Theoretically, at least, the possibility exists that new protein synthesis can play a role. Our "in vivo-in vitro" experiments indicate that estrogens reduce the relative content of DNA and proteins in the ventral prostate of castrated rats. These results agree with the findings of Tunn et al in dogs [28]. If the existence of hyperplasia is reflected in an increase in DNA content [29], then our results indicate an absence of hyperplasia in this acute phase, when estrogen induces increases in prostatic weight. Rather, they suggest an increased rate of cellular loss. In fact, in the E_2 -treated rats, the stroma was less cellular than in the peanut oil-treated rats. Furthermore, in Figures. 1 and 2, the presence of cellular degradation products is evident in the glandular lumina of prostates from

E₂-treated rats but not in tissues from peanut oil-treated or intact animals.

The incorporation of amino acids into proteins (in vivo and in vitro) and thymidine into DNA (in vitro) were higher in the estrogen-treated rats than in the control animals. These results indicate that estrogens also promote new protein synthesis. The participation of new protein synthesis in the acute effects of E₂ on prostatic weight may be related to cellular degradation processes rather than to cellular maintenance or proliferation. Differences in the amino acids uptake in vivo in rats supplemented with E₂ do not account for the observed weight changes (Table VI). The in vitro experiments indicate that tissues derived from E₂-treated animals retain the potential to synthesize proteins (twofold over control values). Despite this, the actual protein synthesis observed in vivo was less than 20% over the control value. Also, no difference was observed in the rates of DNA synthesis in organ culture between explants maintained in 0.01 μ M E₂-supplemented medium and those in a control medium. In addition, when the prostatic regression was held in check (0.1 μ M of E₂), it was found that the thymidine uptake was even lower than that in the control group.

Even though our results in the seven-day culture experiment suggest a direct action of E₂ on the prostate, the in vivo results also point to prolactin involvement. It is possible, therefore, to conclude that, in contrast to recent reports [9], the estrogens in our model have a mixed mechanism of action. The prolactin involvement confirms the impression that, besides a direct effect of estrogens on the prostate, estrogens may also affect the accessory sex glands through changes in prolactin levels [23].

In light of our findings, it seems reasonable to predict the importance of future work on the vascular effects of estrogens in prostatic hypertrophy, and a new understanding of the pathogenesis of benign prostatic hypertrophy and its prevention.

CONCLUSIONS

1. Our results indicate that the estrogen-induced suppression of castration atrophy in Copenhagen rat ventral prostate was mainly the consequence of an edematous condition. Evidence of fluid imbibition or retention was supported by the finding of a significant increase in the ratio of wet/dry tissue weight over the control group.
2. This inhibition of prostate regression could not be accounted for by regeneration through normal growth related macromolecular synthesis (DNA and protein). In fact, estrogen treatment reduced the relative contents of DNA and protein, indicating an increased rate of cellular degradation.
3. Light and electron microscopies showed the presence of fluid in the fibromuscular stroma near the endothelium resulting in gross disorientation of the stromal matrix. The results indicate that fluid imbibition is caused by disturbances of capillary permeability.
4. Partial reversal of the prostatic weight increase and edematization by a prolactin secretion inhibiting agent, taken together with a correlation found between prostate weight and blood levels of prolactin, suggest prolactin involvement in the estrogen effect.
5. Our results confirm a strain difference between Copenhagen and Wistar-Furth rats in terms of sensitivity both in vitro and in vivo. Preliminary findings indicate that prolactin levels in the blood differ greatly between strains.

ACKNOWLEDGMENTS

We wish to thank Dr. J.A. Clemens of Lilly Research Laboratories for performing prolactin measurements, Mr. George Fox for photography and Ms. Ruth Weaver for histology. We also thank Dr. N.H. Altman of Papanicoulou Cancer Research Institute for kindly supplying us with some of the Copenhagen rats. We are grateful to Ms. Anne Agacz and Ms. Liz Glaubitz for technical help and to Ms. Diane Smith for clerical assistance.

This work was supported in part by grants (CA-15436) from the National Cancer Institute (A.A.S.), the US-Spanish Joint Committee for Scientific and Technological Cooperation (J.J.C.), and the Fogarty International Fellowship Award TW 02780 01 (P.A.H.).

REFERENCES

1. Korenchevsky V, Dennison M: The effect of oestrone on normal and castrated male rats. *Biochem J* 28:1474-1485, 1934).
2. Tisell LE: The growth of the ventral prostate, the dorsolateral prostate, the coagulating glands and the seminal vesicles of castrated adrenalectomized rats injected with oestradiol and/or cortisone. *Acta Endocrinol (Copenh)* 68:485-501, 1971.
3. Thompson SA, Rowley DR, Heidger, PM Jr: Effects of estrogen upon the fine structure of epithelium and stroma in the rat ventral prostate gland. *Invest Urol* 17:83-89, 1979.
4. Salander H, Tisell LE: Effects of megestrol on oestradiol-induced growth of the prostatic lobes and the seminal vesicles in castrated rats. *Acta Endocrinol (Copenh)* 82:213-224, 1976.
5. Tisell LE, Andersson H, Angervall L: A morphological study of the prostatic lobes and the seminal vesicles of castrated rats injected with oestradiol and/or insulin. *Urol Res* 4:63-69, 1976.
6. Corrales JJ, Hoisaeter PA, Kadohama N, Murphy GP, Sandberg AA: A model for studies on the response of the ventral prostate to estrogens. *Acta Endocrinol (Copenh)*, 97:125-136, 1981.
7. Cassady JM, Li GS, Spitzner EG, Floss HG, Clemens JA: Ergot alkaloids, ergolines and related compounds as inhibitors of prolactin release. *J Med Chem* 17:300-307, 1974.
8. Li GS, Robinson JM, Floss HG, Cassady JM, Clemens JA: Ergot alkaloids synthesis of 6-methyl-8-ergolines as inhibitors of prolactin release. *J Med Chem* 18:892-895, 1975.
9. Leav I, Merk FB, Ofner P, Goodrich G, Kwan PWL, Stein BM, Sar M, Stumpf WE: Bipotentiality of response to sex hormones by the prostate of castrated or hypophysectomized dogs. *Am J Pathol* 93:69-85, 1978.
10. Frantz WL, Macindoe JH, Turkington RW: Prolactin receptors: Characteristics of the particulate fraction binding activity. *J Endocrinol* 60:485-497, 1974.
11. Aragona C, Friesen HG: Specific prolactin binding sites in the prostate and testis of rats. *Endocrinology* 97:677-684, 1975.
12. Boyns AR, Cole EN, Golder MP, Danutra V, Harper ME, Brownsey B, Cowley T, Jones GE, Griffiths K: Prolactin studies with prostate. In Boyns AR, Griffiths K (eds): "Prolactin and Carcinogenesis." Cardiff, Wales: Alpha Omega Alpha, 1972, pp 200-206.
13. Manandhar MSP, Thomas JA: Effect of prolactin on the metabolism of androgens by the rat ventral prostate gland in vitro. *Invest Urol* 14:20-22, 1976.
14. Kadohama N, Kirdani RY, Murphy GP, Sandberg AA: 5 α -Reductase as a target enzyme for anti-prostatic drugs in organ culture. *Oncology* 34:123-128, 1977.
15. Burton K: A study of the conditions and mechanism of the diphenylamine reaction for the colorimetric estimation of deoxyribonucleic acid. *Biochem J* 62:315-323, 1956.
16. Lowry OH, Rosenbrough NJ, Farr AL, Randall RJ: Protein measurement with folin phenol reagent. *J Biol Chem* 193:265-275, 1951.
17. Millonig G: Advantages of a phosphate buffer for OsO₄ solution in fixation. *J Appl Physics* 32:1637-1661, 1961.
18. Reynolds ES: The use of lead citrate at high pH as an electron-opaque stain in electron microscopy. *J Cell Biol* 17:208-213, 1963.
19. Huggins C, Clark PJ: Quantitative studies of prostatic secretion. II. The effect of castration and of

- estrogen injection on the normal and on the hyperplastic prostatic glands of the dog. *J Exp Med* 72:747-761, 1940.
20. Spaziani E: Accessory reproductive organs in mammals: Control of cell and tissue transport by sex hormones. *Pharmacol Rev* 72:207-286, 1975.
 21. Roberts S, Szego CM: Steroid interaction in the metabolism of reproductive target organs. *Physiol Rev* 33:593-629, 1953.
 22. Spaziani E, Szego CM: Early effects of estradiol and cortisol on water and electrolyte shifts in the uterus of the immature rat. *Am J Physiol* 197:355-359, 1959.
 23. Negro-Vilar A: Prolactin and the growth of prostate and seminal vesicles. In Spring-Mills E, Hafez ESE (eds): "Male Accessory Sex Glands." Elsevier/North-Holland Biomedical Press, 1980, pp 223-233.
 24. Müntzing J, Kirdani R, Murphy GP, Sandberg AA: Hormonal control of zinc uptake and binding in the rat dorsolateral prostate. *Invest Urol* 14:492-495, 1977.
 25. Resnick MI, Grayhack JT: Effect of 2-bromo- α -ergocriptine (CB-154) on estrogen-induced growth of the rat prostate. *Invest Urol* 15:392-394, 1978.
 26. Negro-Vilar A, Saad WA, McCann SM: Evidence for a role of prolactin in prostate and seminal vesicle growth in immature male rats. *Endocrinology* 100:729-737, 1977.
 27. Dunning WF: Prostate cancer in the rat. *Natl Cancer Inst Monogr* 12:351-369, 1963.
 28. Tunn V, Senge T, Schenck B, Neumann F: Biochemical and histological studies on prostates in castrated dogs after treatment with androstenediol, estradiol and cyproterone acetate. *Acta Endocrinol (Copenh)* 91:373-384, 1979.
 29. DeKlerk DP, Heston WDW, Coffey DS: Studies on the role of macromolecular synthesis in the growth of the prostate. In Grayhack T, Wilson JD, Scherbenske MJ (eds): "Benign Prostatic Hyperplasia." DHEW Publ, 1975, pp 43-54.